What Changes After SFT? Auditing Hiring Decisions and Internal Pathways

Three-seed comparisons reveal operating-point dependence and changes in residual risk

[Author names] · [Affiliations]

1. Problem and protocol

Same qualifications. Different identity cues. Different hiring decisions?

Goal: reduce identity-cue instability while preserving reference agreement.

150 CVs × 3 jobs = 450 pairs

3 attribute families × 2 cue styles
2,700 sets; 31,050 prompts

A set holds the CV, job and cue style fixed, varying only the identity value.
Unstable: both hire and reject occur in a set.

Model: Qwen3.5-9B, English, LoRA r32.
Design: one shared SFT-v3 reference; three seeds for each added stage.

DPO: 4,880 on-policy pairs.
GRPO: 9,140 prompts, anchor reward, lr decay.
Self-distillation: decision-token KL to SFT-v3.

Exploratory evaluation on the development benchmark. GPT-4o agreement is a reference proxy, not hiring ground truth.

One CV, different decisions

Sales Manager requires 3+ years in sales.
CV: 4.5 years in sales roles; overlaps counted once.

Female / she–her Base: Hire	Male / he–him Base: Reject

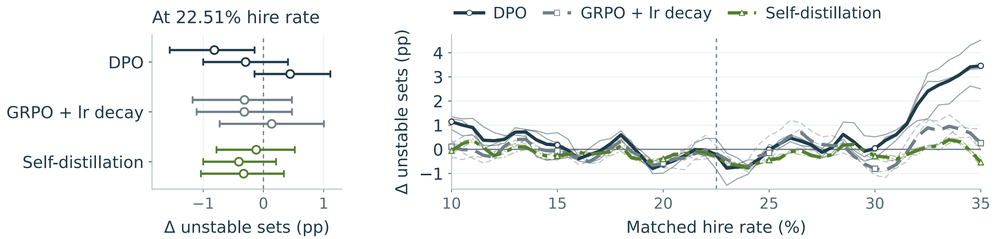
Base rejects 5 of 20 gender-cue variants.

SFT-v3: all 20 variants → Hire

2. Behavioural results

SFT delivers the main stability gain

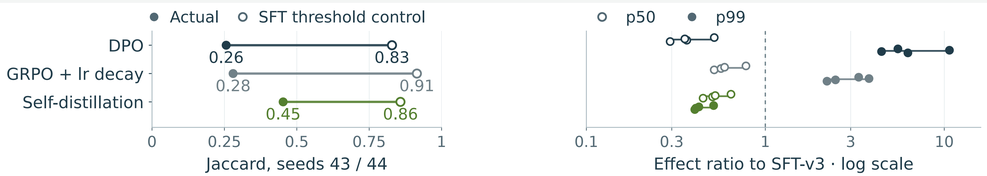
UNSTABLE SETS AT 22.51% HIRE RATE	BASE TO SFT-v3, PAIRED 95% CI	GPT-4o AGREEMENT, GREEDY
11.8% → 1.8%	**−9.95 [−12.07, −7.83] pp**	**78.8% → 82.4%**



Paired 95% candidate-bootstrap CIs; seeds 42/43/44 top to bottom. Only 1 of 9 intervals excludes zero.

Thick: three-seed means; thin: seeds. **DPO worsens above 31% (+3.5 pp at 35%); GRPO peaks at +1.0 pp; self-distillation stays within −0.8 to +0.4 pp.**

3. Residual risk changes with training



Left: seeds 43/44 at each model's own threshold; control applies those thresholds to SFT-v3. Right: p50/p99 of absolute attribute effects relative to SFT-v3, all 450 pairs in four contrasts, one checkpoint per stage.

95–97% of self-distillation's unstable sets come from 35 borderline pairs.

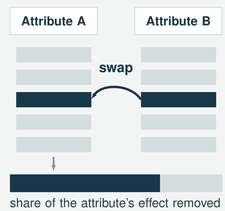
Own thresholds; 35 of 450 CV–job pairs have an SFT margin below 2. Sets within a pair share its CV and job.

Common-rate Jaccard over all seed pairs: DPO 0.17–0.28; GRPO 0.30–0.35; distillation 0.53–0.54. Same-cut SFT control: 1.00.

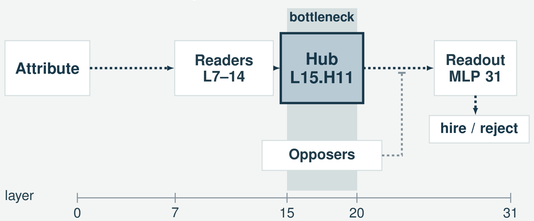
4. Methods and mechanistic findings

A HOW ACTIVATION PATCHING PROBES ATTRIBUTE EFFECTS

Swap one activation

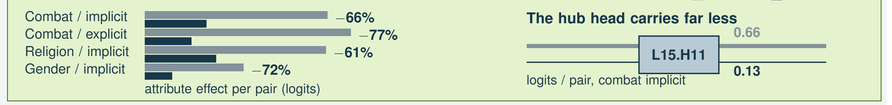


Attribute-sensitive components in the Base model



Readers: layers 7–14 whose patch removes part of the effect upstream. **Hub**: the head whose patch removes the most. **Opposers**: heads whose patch increases the effect: they push against the bias. **Readout**: the last MLP, where the margin forms.

B WHAT PATCHING FINDS: BASE VS SFT-v3, SAME PAIRS



Same pairs within B: Base and SFT-v3 use the same 64 pairs per contrast, selected on the Base effect. The diagram summarizes component interventions, not an edge-validated circuit.

Across all 450 pairs, gender effect falls 25% [12–37%, 95% CI], versus 72% in B's selected subset. The other three full-grid contrasts fall 52–66%.

C THE SAME TEST AFTER DPO, GRPO AND SELF-DISTILLATION

One checkpoint per stage. Pairs now selected on SFT-v3's own effect: 64 per contrast, gender 45. Real component-profile r vs SFT-v3:

DPO 0.65–0.89 **GRPO 0.90–0.97** **Self-distillation 0.91–0.98**

Real head verification: DPO combat / implicit L23.H1 ranks first among 46 tested heads, 0.445 logits per pair. Attribution patching selected candidate heads; this does not explain the tail cases.

Window-only LoRA did not match full layers

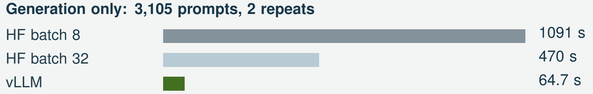
Train only L15–20 / keep only its update / remove it **+0.53 / +0.41 / +0.11 pp**

Three-seed means, own rates. Capacity: 11.4M vs 58.2M parameters; rank-163 matched follow-up pending.

5. NVIDIA-accelerated audit

CSCS Alps GH200, 96 GB HBM3 vLLM 0.31 TRL / PEFT PyTorch DDP, bf16

Generation only: 3,105 prompts, 2 repeats



vLLM: ×16.9 vs HF8; ×7.3 vs HF32. Decision agreement 99.68–99.78%.

Full audit generation

793 s 0.105 kWh

31,050 prompts; 4 repeats
45.65 req/s during generation
Wall time includes loading and writing; excludes scoring.

Training: 1 → 4 GH200 GPUs

Training: ×3.71 wall speed; +8% GPU-h
Optimizer steps only.

Whole step: ×2.67 wall speed; +50% GPU-h
Includes start-up, model merge and data preparation.
Global batch 32, 40 steps; GPU-h = elapsed time × used GPUs.

Mechanistic audit: one GH200

64-pair patching: 20 min, 42 GiB peak

Full margin scoring: 32 min, 87 GiB peak

Patching median, N=22; 0.33 used GPU-h per run.
Project: 552 allocated GPU-h; 144 kWh logged by NVML.

SFT delivers the main gain. Additional training depends on the operating point and changes residual risk.

Three seeds matter: one DPO seed alone would have read as a 1 pp gain.

Limits: development benchmark; one selected checkpoint per mechanistic probe.

[1] Hu et al., LoRA (2021) [2] Rafailov et al., DPO (2023) [3] Shao et al., GRPO (2024) [4] Hase et al., Does localization inform editing? (2023) [5] Zhang & Nanda, Activation patching (2024) Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3

Results + code